

Lokesh Venkatesh

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Research Interests: **Complex Systems** | **Computational Neuroscience** | **Machine Learning**



EDUCATION

Indian Institute of Science Education and Research (IISER), Pune <i>Integrated BS–MS Natural Sciences — Interdisciplinary coursework across the basic sciences</i>	Nov 2022 — Present
Yuvabharathi Public School, Coimbatore <i>AISSCE Grade 12 (CBSE) — Physics, Mathematics, Chemistry, Informatics Practices, English</i>	Jul 2022
Yuvabharathi Public School, Coimbatore <i>AISSE Grade 10 (CBSE) — Science, Mathematics</i>	Jul 2020

RESEARCH EXPERIENCE

Tumour-immune microenvironment interactions in cancer progression <i>Advisors: Prof. Ram Rup Sarkar, NCL Pune & Dr. Chandrakala Meena, IISER Pune</i> Carrying out fixed-time simulations of an ODE model with further parameter & bifurcation analyses to investigate how tumour-immune microenvironment interactions influence the dynamics of cancer progression.	Nov 2025 — Present
Self-supervised Representation Learning for Circadian Time Series <i>Advisor: Dr. Shaon Chakrabarti, NCBS Bangalore</i> - Implemented representation learning algorithms like the variational autoencoder architecture on patient actigraphy datasets, followed by clustering, in order to obtain their learned low-dimensional representations. - Analysed the learnt embeddings to identify the respective circadian phases and isolate what specific variables carry the most significance in the multivariate time series.	May 2025 — Jul 2025
Study of the Mitochondrial Genome using Motif Distributions <i>Advisor: Prof. MS Madhusudhan, IISER Pune</i> - Analysed motif distributions across a database of mitochondrial genomes as a metric to both reconstruct phylogenetic relationships over the course of its evolution, as well as identify genomic translocations. - Also looked at possible regulatory correlations between mitochondrial & nuclear genome promoter sequences.	Aug 2024 — Apr 2025

ACADEMIC PROJECTS & TRAINING

SCHOOLS & WORKSHOPS

Spins, Games & Networks, 2025 Two-week fully-funded winter school on theoretical & computational approaches to complexity science at IMSc Chennai	Dec 2025
NeuroAI Ten-day interactive online course on principles connecting artificial & biological neural nets by the Neuromatch Academy	Jul 2025
Brain, Dynamics & Computation, 2025 Two-week fully-funded summer school on network science-approaches to computational neuroscience at IMSc Chennai	May 2025

COURSE & READING PROJECTS

Tools for Complexity Science — Reading Project Undertaking a reading project (<i>Advisor: Dr. Chandrakala Meena</i>) to study the various methods & tools, largely based on Stefan Thurner et al's book, used to model complex biological and social systems' phenomena.	Sep 2025 — Present
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RELEVANT COURSEWORK

Data Science — Applied Mathematical Methods, Deep Learning, Image & Video Processing, Mathematical & Statistical Optimisation, Computational & Functional Genomics, Signal Processing, Mathematical Finance

Physics — Classical Mechanics, Statistical Mechanics, Nonlinear Dynamics, Physics of Biological Systems

Mathematics — Multivariable Calculus, Linear Algebra, Probability, Graph Theory, Markov Processes

Biology — Systems Biology, Neurobiology, Bioinformatics, Cellular Biophysics I & II, Animal Behaviour, Ecology & Evolution, Cell Biology, Physiology, Genetics

Humanities & Social Sciences — Science & the World, Philosophy of Science, Cognitive Bases of Science, Science & Society, Public Policy I & II

SKILLS

Computational: Numerical simulation of ODE systems, Parameter exploration and bifurcation analysis, Time-series analysis, Representation learning algorithms, Bioinformatics Tools

Technical: High-performance computing, Large-scale data processing, Pipeline development

Programming: Python (NumPy, Pandas, SciPy, PyTorch, Scikit-learn, Biopython), Julia, Bash, Markdown, $\text{\LaTeX}$

Languages: English, Tamil (bilingual proficiency); Hindi (working proficiency)

SCHOLARSHIPS & AWARDS

INSPIRE's Scholarship for Higher Education (SHE)

INSPIRE-SHE Scholar (4 Lakh INR $\approx$ 4,500 USD) - awarded to top 0.1% (approx. 10,000) students nationwide by the Department of Science and Technology, Government of India

EXTRACURRICULAR & OUTREACH

Academic Support Group, IISER Pune

Aug 2023 — Sep 2024

Senior teaching & logistics volunteer for a student collective mentoring socio-economically underprivileged students

Kalpa — IISER Pune's Student Media House

Mar 2023 — Jan 2024

Member of the student media body's core team that coordinated investigative journalism projects on campus matters

REFERENCES

Prof. M.S. Madhusudhan (IISER Pune)

Email: MADHUSUDHAN@IISERPUNE.AC.IN

Dr. Chandrakala Meena (IISER Pune)

Email: CHANDRAKALA@IISERPUNE.AC.IN

Dr. Shaon Chakrabarti (NCBS Bangalore)

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